

1. What Are LLM Leaderboards and Why Do They Exist?

The Core Definition of LLM Leaderboards

Before we begin, let us establish what **LLM Leaderboards** actually are. You have likely seen them or may even use them in your day-to-day work or inside your company. To understand leaderboards, we must first understand **benchmarks**.

- **Benchmarks** are essentially exams designed to test Large Language Models (LLMs) on specific, individual capability dimensions.
- Once these models take these exams, we get results. We need a place to publish, compare, and display these results. That centralized space is the leaderboard.
- An **LLM Leaderboard** is a **public ranking system or comparison table** that shows how different LLMs perform on a common set of evaluations.

By aggregating these benchmark results, a leaderboard becomes a single, centralized destination. Instead of navigating dozens of different sites, we can compare multiple models from different creators on a single benchmark in one place, allowing us to immediately see which model is performing best on that specific test.

Why Do LLM Leaderboards Exist?

There are several core reasons why these platforms are built and why we rely on them:

- **Comparing Models Across Labs with a Common Reference:** Different artificial intelligence research labs (such as OpenAI, Anthropic, or Google) train their own models. Leaderboards allow us to evaluate all of these models using the exact same exam as a common reference point. It shows us who took the top spot, who placed second, and who ended up last, helping us determine which model is the most capable.
- **Providing Trust and Objectivity via Third-Party Verification:** If OpenAI or Anthropic self-reports that their new model scored extremely high on a specific benchmark, we might be skeptical. These labs have high financial and reputational stakes, so they naturally want to present their models in the best possible light. However, leaderboards are generally hosted by **independent third parties**. Because these third parties do not have a vested interest in the success of any single model, we can place much higher trust in their objective, side-by-side evaluations.
- **Guiding Model Selection When Resources are Limited:** Imagine you want to build a math chatbot for students. In an ideal world, you would test every single model yourself to find the best one. However, there are hundreds of models in existence today. It is practically impossible for an individual developer or a small company to test all of them. Doing so would consume an immense amount of money, time, and computational effort. Because leaderboards run these

evaluations for us, our job is simplified: we can go to the leaderboard, view the top performers, and select our candidates.

- **Identifying Benchmark Saturation:** Leaderboards allow us to see when a benchmark is no longer useful because it has **saturated**. If we look at a leaderboard and observe **clustering**—where the top ten models are all scoring within a tiny, compressed range, such as 92% to 94% on a general knowledge capability test—we can immediately tell that the benchmark is saturating. It is no longer difficult enough to distinguish the performance differences between top-tier models.
 - **Discovering New and Cost-Effective Models:** While the top three or four spots on a leaderboard are usually dominated by the major proprietary models from Google, OpenAI, and Anthropic, scrolling down to ranks 10, 12, 15, or 20 reveals lesser-known models. While these models might not be the absolute best in terms of raw accuracy, they are highly valuable because they are **generally much cheaper** and may be perfectly sufficient for your specific application.
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2. The Five Key Stakeholders of LLM Leaderboards

Leaderboards are not just used by developers; they serve multiple distinct groups in the AI ecosystem. Let us look at how each stakeholder interacts with them:

1. AI Engineers (Our Group)

- **Our Purpose:** We use leaderboards for **shortlisting and filtering candidate models**. When building an LLM-powered application, we do not simply choose the number-one ranked model on a leaderboard and write our code. Instead, we use the leaderboard as a filter to narrow down hundreds of available models to a shortlist of 3 to 5 candidate models. Once we have this shortlist, we run our own custom evaluations on them to identify the single best model for our specific application. The leaderboard helps us perform the massive initial filter from 100 models down to 5.

2. Frontier Labs

- **Their Purpose:** Large model-building companies (like OpenAI, Google, and Anthropic) use leaderboards to benchmark their competitive positions and **plan and strategize model releases**.
- For example, if I am OpenAI and my current model is **GPT 5.5**, and a competitor has **OPS 4.8** leading the benchmarks, and the new iteration of my model currently in training fails to beat OPS 4.8, I will not release it. Releasing a model that performs worse than a competitor's existing model would be disastrous marketing, and the community would immediately criticize it. I will instead continue training and only release the model when it significantly outperforms the competition.
- **Stealth Mode Testing:** Frontier labs also use leaderboards for stealth marketing. They often

submit unreleased models to public leaderboards under **hidden, funny pseudonyms** (such as "Nano Banana" on an image evaluation leaderboard). They do not disclose that Google or any other major company is behind the model. Once this mystery model dominates the benchmarks and generates organic hype, the lab officially reveals its identity and releases the model.

3. Researchers

- **Their Purpose:** Researchers study leaderboards to observe macro trends in the field, such as which benchmarks are saturating and where performance gaps still exist. This helps them identify new techniques, generate fresh ideas, and define new **research directions**.

4. Policy Makers and Safety Institutes

- **Their Purpose:** Government and regulatory bodies monitor leaderboards to track the state of AI capabilities. If a new model emerges that suddenly and drastically leaves all other models behind, they must step in to evaluate safety risks. This occurred with "**Fable Five**", where the US Government intervened because the model's capabilities appeared potentially dangerous.

5. The Open Source Community

- **Their Purpose:** For small research labs consisting of only 10 to 15 people, leaderboards are the ultimate tool for **discovery, marketing, and publicity**. If a small lab trains a highly optimized model that achieves a top-five position on a major benchmark, their model instantly goes viral. This is exactly how several prominent Chinese research labs emerged overnight—by releasing models that suddenly rivaled top-tier proprietary models on specific leaderboard benchmarks, gaining immediate global recognition and adoption.
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3. The Four Types of LLM Leaderboards

When you search the internet, you will encounter four distinct types of leaderboards. Let us examine each category, its purpose, and its limitations:

1. Benchmark Specific Leaderboards

- **How They Work:** These leaderboards rank models based entirely on the results of a **single, specific benchmark exam** (such as **MMLU** for general knowledge, **HumanEval** for coding, **GSM8K** for mathematics, or **GPQA**).
- **Core Objective:** To show which model performs best on that one specific test.
- **The Limitation:** They provide an extremely **narrow, one dimensional view**. You only learn how a model performs on that specific exam, but you gain absolutely no understanding of its overall general performance or capabilities in other domains.
- **Real World Example:** The **HLE (Humanities Last Exam)** leaderboard. This is an exam specific leaderboard that displays scores (such as **Gemini 3 Pro** scoring at **38%**) alongside

specific evaluation details like **calibration error**. These leaderboards are usually built and maintained directly by the research teams who created the benchmarks. They are generally not the most useful for overall model selection.

2. Multi Benchmark Leaderboards

- **How They Work:** Rather than relying on a single test, these leaderboards evaluate models across **multiple benchmarks and evaluation dimensions** (e.g., combining reasoning, coding, mathematics, instruction following, and data analysis) to calculate a **cumulative or overall score**.
- **Core Objective:** To provide a comprehensive, multi-dimensional view of a model's general capabilities across diverse tasks.
- **Operational and Performance Metrics:** The most useful multi-benchmark leaderboards provide highly valuable operational data alongside capability scores, including:
 - **Cost per token**
 - **Latency**
 - **Output speed**
 - **Context window size**
- **Real World Examples:**
 - **LiveBench:** A contamination-free benchmark containing **23 objective tasks across 7 categories** (such as reasoning, coding, and agentic coding). It displays individual category scores alongside an overall aggregated score to give a clear picture of a model's general capabilities.
 - **Artificial Analysis:** A dedicated platform that specializes in building highly detailed, product-grade leaderboards. They provide an exhaustive collection of individual tables for different dimensions—such as an **Intelligence leaderboard**, a **Speed leaderboard**, a **Cost per task leaderboard**, and specific benchmark tables like **HLE** and **GPQA Diamond**. They also host specialized leaderboards for **coding agents, speech, image, audio, and hardware**, alongside a comprehensive, aggregated leaderboard that combines everything.
- **My Verdict:** This is the **most useful and practical category** of leaderboards. This is the type that I use the most, and it is what you will rely on to find which model offers the strongest combination of capability, cost, and speed.

3. Human Preference Based Leaderboards

- **How They Work:** These leaderboards do not rely on traditional automated benchmarks. Instead, they rank models using **direct human comparison and voting**.
- When a user visits the platform, they enter a prompt of their choice. Behind the scenes, two anonymous models (Model A and Model B) generate responses to that exact prompt. The user

reads both responses and votes on which one is superior based on qualitative criteria like **helpfulness, clarity, writing quality, or creativity**. Once a massive volume of these blind votes is collected globally, the platform aggregates them to compile a dynamic ranking.

- **Real World Example: LMSYS Chatbot Arena (LM Arena).** In its "battle mode", a user can ask a question (such as *"What are LLM leaderboards?"*), and two unidentified models generate answers. After evaluating them, the user selects "A is better", "B is better", "both are good", or "both are bad". After voting, the system reveals the true identities of the models (such as revealing **Claude OPS 4.8 Search** and **Fable**). These votes compile into a live ranking featuring top models (such as **Fable 5, 5.6, and Sol 4.8 Thinking**) across specialized sub categories like code, image, and video.
- **The Limitation: Human Bias.** Human evaluators are often swayed by superficial factors rather than objective correctness. We tend to vote for answers that are beautifully formatted, highly confident, polite, or tailored to our personal preferences, even if the underlying information is less accurate than a plain-text, objectively correct response.
- **My Verdict:** Despite the bias, at a global scale with millions of votes, the data becomes highly reliable. This is proven by the fact that the objectively strongest models still dominate the top of these rankings. These leaderboards are highly popular and heavily leveraged for marketing.

4. Application-Specific Leaderboards

- **How They Work:** These are highly specialized leaderboards built around a **single, dedicated domain or specific workflow**. They aggregate multiple benchmarks relevant to that specific task into a unified scoring system.
 - **Examples:** Dedicated leaderboards for coding tasks, agentic workflows, SQL query generation, or medical question-answering.
 - **Real-World Example:** The **Berkeley Function Calling Leaderboard (BFCL)**. This platform focuses entirely on evaluating and ranking a model's **tool-calling and function-calling capacity**. While it might utilize multiple distinct benchmarks under the hood, it operates strictly within this single operational domain.
 - **My Verdict:** These are highly valuable if you are building an application in that specific niche.
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4. Why You Cannot Blindly Trust LLM Leaderboards

As AI engineers, you must overcome the cognitive bias that says, *"If a model is ranked number one on a leaderboard, it must be the best choice for my application."* Leaderboards are incredibly powerful, but blindly trusting them is a massive mistake. Let us examine the major reasons why:

1. Benchmark Performance May Not Transfer to Real Applications

- A model scoring 80% or 90% on a clean benchmark does not guarantee it will perform well in

your production environment.

- **The Kaggle Analogy:** In data science, solving problems on Kaggle does not automatically make you a great real-world data scientist. Why? Because Kaggle provides perfectly cleaned datasets and highly explicit problem statements. In the real world, data is incredibly messy and chaotic.
- Similarly, benchmark datasets are highly curated and cleaned. Real world user interactions are filled with **ambiguous requests, missing information, company specific contextual data, tool failures, and highly unusual edge cases**. We cannot know from a leaderboard score alone how a model will handle these chaotic real world variables.

2. Benchmark and Leaderboard Contamination

- We have discussed how easily benchmarks get **contaminated**. If a benchmark's evaluation data is accidentally or intentionally included in a model's training pre-training or fine-tuning datasets, the model will essentially "memorize" the answers.
- When tested, the model will output an artificially **inflated and non genuine score** because it has already seen the questions, rendering its high leaderboard ranking completely meaningless.

3. Models Over Optimized for Leaderboards (Goodhart's Law)

- When a specific leaderboard (like LMSYS Chatbot Arena) becomes highly popular and serves as the primary metric for public prestige, model-building companies focus their entire engineering effort on topping that specific leaderboard.
- Since humans vote on these platforms, companies fine tune their models to output highly formatted, overly polite, and conversational answers designed specifically to please human evaluators, rather than focusing on improving core reasoning or accuracy.
- **Goodhart's Law:** This phenomenon perfectly exemplifies Goodhart's Law: "**When a measure becomes a target, it ceases to be a good measure.**"
- **The Car Mileage Analogy:** Imagine a car manufacturer in India, a market where consumers heavily prioritize fuel efficiency (mileage). If the company's engineering team focuses 100% of their energy on optimizing mileage, they will successfully build a car with incredible fuel economy. However, because they ignored everything else, the final car's driving dynamics, safety, and 0-to-100 acceleration will be terrible. The overall vehicle is poor, even though it excels at the single targeted metric.
- The same is happening in AI: models are being fine-tuned specifically to climb leaderboards, resulting in high scores but stagnant real-world utility.

4. Lack of Transparency in Composite Leaderboards

- Many multi-benchmark leaderboards combine several tests to produce a single composite score, but fail to disclose critical details:
 - Which specific benchmarks are included or excluded?

- How are different capabilities weighted?
- What normalization methods were used to merge the scores?
- This lack of transparency hides critical performance discrepancies and can mislead you if a model's high average score is driven by a capability that is irrelevant to your application.

5. Rank Bias and Meaningless Marginal Differences

- Engineers often fall victim to **rank bias**, automatically preferring a model ranked 3rd over a model ranked 5th, even if the actual score difference is trivial (e.g., **84.3% vs. 84.1%**).
- **The IIT-JEE Analogy:** In highly competitive exams like the IIT-JEE, the academic difference between Rank 1 and Rank 25 is practically negligible it often comes down to a simple grading variation or a simple careless mistake on one or two questions. They are effectively at the exact same intellectual level.
- Similarly, two models with a 0.2% score difference on a leaderboard are functionally identical. In fact, the 5th-ranked model might actually perform significantly better, faster, or cheaper for your specific application, but rank bias might prevent you from even considering it.

6. Human Biases in Preference Evals

- In human-voted leaderboards, shorter, direct, and objectively correct answers are often penalized in favor of longer, highly confident, beautifully formatted, or entertaining responses. An objectively inferior model that has mastered stylistic formatting can easily outrank an objectively superior model that output a blunt but correct answer.

7. Stale, Incomplete, or Self-Reported Data

- Leaderboards are frequently **outdated**. For example, the Humanity's Last Exam leaderboard still lists Gemini 3 Pro but lacks the most recent models like Fable or 5.6 Sol because the maintainers have not updated it.
- You will often see outdated model versions, discontinued models, or rankings that rely entirely on **self-reported data** submitted directly by the model creators themselves, which defeats the purpose of objective evaluation.

5. Step-by-Step Guide: How to Correctly Use LLM Leaderboards as an AI Engineer

To ensure you use leaderboards effectively without falling into these traps, you must follow this structured, five-step methodology:

Step 1: Clearly Define Your Application Constraints First

Before you open a single leaderboard website, write down your application's exact operational

requirements. You must define:

- **The Type of Application** (e.g., math solver, document search, chatbot).
- **Latency Limits** (how fast must the model respond?).
- **Cost Tolerances** (what is your budget per thousand tokens?).
- **Context Window Requirements** (how much text must the model process at once?).
- **Deployment Constraints** (can you use public APIs like Claude, or are you legally/operationally mandated to run an **on-premise/local** model?).
- *Why this matters:* Establishing these constraints upfront prevents rank bias. For instance, if you require a strict on-premise deployment, you immediately rule out proprietary models like Claude Fable and focus your search entirely on highly capable open-source models, keeping your evaluation objective.

Step 2: Navigate to the Correct, Domain Specific Leaderboard

Do not waste time looking at generic, single benchmark tables. Match your application to the correct platform:

- If you are building a **coding agent or complex tool use system**, go to a tool calling or coding specific leaderboard (like the **Berkeley Function Calling Leaderboard**).
- If you are building a **conversational chatbot**, look at human preference tables like the **LMSYS Chatbot Arena**.
- If you are building a **RAG (Retrieval-Augmented Generation) pipeline**, you must select an embedding model first. Therefore, go to the **MTEB (Massive Text Embedding Benchmark)** leaderboard, which ranks embedding models.
- If you have strict **budget or speed constraints**, look at operational leaderboards like **Artificial Analysis** or **Vellum** to get precise, real time data on inference speed, cost, and latency.

Step 3: Analyze the Leaderboard Details and Read the Fine Print

Do not just look at the raw ranking number. Read the definitions, the FAQs, and the "star marks" to evaluate the leaderboard's integrity:

- What exactly is being measured, and how is it scored?
- Who ran the evaluations?
- What was the inference budget allowed for the models?
- Did the evaluation involve reasoning steps?
- How old is the evaluation dataset, is it updated, and is a private test set used to prevent contamination?
- Are the benchmarks saturated (showing clustered scores)?

- Is there a **confidence interval** provided for close scores? (If scores fall within the interval, treat the models as identical).
- For composite leaderboards, what are the weights assigned to individual capability scores?

Step 4: Shortlist a Pool of 3 to 5 Candidate Models

Based on your application criteria and your leaderboard analysis, filter the massive list of models down to a shortlist of **3 to 5 candidate models**. Do not try to pick a single winner yet.

Step 5: Run Your Custom Internal Evaluations

This is the final and most important step. Take your shortlist of 3 to 5 models and run them against your own **custom evaluation dataset** containing real world prompts, messy data, and edge cases specific to your business. The model that performs best on your internal evaluation is your final choice.

6. The Golden Rule of LLM Leaderboards

If there is one absolute takeaway you must remember from this session, it is this:

"Leaderboards are a filtering tool, not a selection tool. They are not a decision tool."

You never go to a leaderboard to make your final decision on which model to deploy. You use leaderboards solely to filter 100+ models down to a shortlist of 3 to 5, and then you let your **custom internal evaluations** make the final decision.